

Mixed Reality and Real-Time X-Ray Simulation in Vet Radiography Training: A User-Centered Comparative Study

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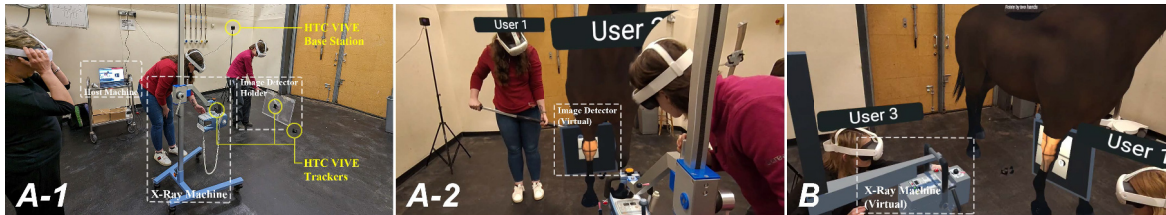


Figure 1: Students in a group taking X-ray images collaboratively under the guidance of an on-site lecturer in physical-equipment-based MR system (Fig. 1-A-1). First-person view of the lecturer in this system (Fig. 1-A-2). Third-person view of two radiography specialists taking X-ray images collaboratively using virtual equipment in virtual-equipment-based MR system (Fig. 1-B).

ABSTRACT

Proficient training in Diagnostic Imaging (DI) is crucial for veterinary students. However, the training process faces challenges, including limited access to animals and strict radiation safety regulations. Can Mixed Reality (MR) address these issues? Two co-located collaborative MR teaching systems were implemented to simulate the horse DI procedure with real-time X-ray simulation. One system integrated actual X-ray equipment, while the other was entirely virtual. A user-centered comparative study was conducted, assessing students' performance and opinions to explore the differences and the necessity of physical versus virtual equipment. The performance of 22 veterinary students and four DI experts was evaluated, focusing on the quality of the radiographs and the time taken to obtain the X-ray images. Students initially underperformed compared to experts using physical equipment but matched expert performance by the final image, while they performed similarly with virtual equipment. Experts took X-rays faster than students in both cases. Both systems offered comparable user experiences, though most preferred virtual equipment for its usability and presentation. Physical equipment remained essential for realistic simulation. The results show that the MR system, combined with real-time X-ray image synthesis, has great potential to overcome obstacles in veterinary DI training and serve as a support tool for both training and assessment.

Index Terms: Mixed Reality, Hybrid Reality environment, Co-location, Tangible user interface, Radiography.

1 INTRODUCTION

In DI education, introducing radiographic skills during the pre-clinical phase is critical for veterinary students. However, training students in radiographic techniques for equine patients presents significant challenges. These include the large size of these animals, limitations in patient availability, and ethical concerns. Consequently, the current DI institution lacks a standardized assessment method, and conventional lecturing often fails to provide comprehensive, hands-on radiographic experience [40]. As a result, newly

graduated students may require additional X-ray images to achieve the correct radiographic view, increasing radiation exposure for both the patients and the operating room team. Researchers have attempted to use low-fidelity models to simulate the process [17]. However, the simplified nature of these simulators limits their ability to replicate detailed aspects of radiographic processes and does not emulate the actual weight of real X-ray equipment, thereby restricting the realistic handling experience. Simulating the entire process in a virtual environment offers a solution to these challenges. By using head-mounted-display (HMD) based Virtual Reality (VR), Augmented Reality (AR), and MR, collectively known as xReality [21], these technologies have been shown to support medical education [38] and enhance human radiography skills interactively and collaboratively [1, 18, 20, 25]. Students trained in virtual environments display superior performance, particularly in X-ray techniques. According to previous research, the collaborative co-located MR system effectively improves the teaching of veterinary radiation safety rules by providing an immersive and collaborative learning experience [40]. This fully virtual system combined with real-time X-ray image simulation is termed Virtual Equipment Mixed Reality (VEMR) in this study. Real-time radiograph simulation provides an interactive teaching environment that optimizes equipment usage, reduces procedural time, and improves trainee performance [13, 15, 29, 30, 33]. Seamless integration and real-time synchronization of physical and virtual interactions via tangible interfaces can enhance the sense of presence in virtual environments, particularly for tasks that simulate real-world activities [4, 19, 34, 35]. These enhancements could benefit DI training.

This study introduces a tracker-based, co-located collaborative MR system designed to replicate DI procedures by tracking physical X-ray equipment annotated with virtual contexts (Fig. 1-A-2), which is referred to as the Physical Equipment Mixed Reality (PEMR) system in this study. The system generates real-time simulated X-ray images based on real-time positioning. The expert review study showed that radiology experts found this system notable and engaging, with its usefulness highlighted for providing a safe and repeatable learning experience [39]. This is especially beneficial for novice trainees who require extensive hands-on practice.

To assess the system's usefulness to students and explore the necessity of using physical equipment in the simulation, a user-centered comparative cross-sectional study was conducted to evaluate PEMR (Fig. 1-A) and compare it with VEMR (Fig. 1-B). The

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independent variable in this study is the involvement of physical equipment, while the dependent variables are the students' performance measured by their scores and the time taken to complete a DI procedure and their behavior measured by their movement data. The contributions of the study are as follows: (i) Evaluation of the proposed system in terms of perceived workload, user behavior, and student performance compared to experts. (ii) Comparison of the proposed system with a fully virtual interaction system. (iii) Collection of feedback and insights regarding the future application of real medical equipment in the MR environment.

2 RELATED WORK

2.1 Diagnostic Imaging Training in Virtual Environment

Virtual simulation has emerged as a cost-effective alternative to traditional laboratory methods, offering enhanced student satisfaction and performance in human X-ray diagnostic techniques [6, 9]. Previous research highlights the advantages of VR in visualizing radiation fields, aiding trainees in avoiding high-radiation areas and safely adjusting practices without exposure to real X-rays [8, 10, 28]. The integration of HMDs and dynamic interactions enables users to interact more realistically with medical imaging training environments where trained individuals outperform novice radiographers in positioning tasks [25] and report high levels of satisfaction [26]. These VR simulations have proven valuable in teaching medical imaging positioning skills and are recognized as safe and effective pedagogical tools [1, 18, 20]. Furthermore, immersive VR simulations provide interactive learning experiences, often receiving higher scores compared to 2D desktop systems [18, 24, 32]. These findings underscore the potential of VR for skill acquisition and engagement in human medicine, paving the way for exploration in veterinary radiology, where similar techniques are rarely applied.

While immersive VR enables collaborative learning, it may also isolate users from their physical surroundings, potentially hindering communication [2]. In contrast, MR environments can address this limitation of immersive VR and be adapted for veterinary radiography training, enabling more interactive and collaborative educational experiences without the risks associated with radiation exposure [37]. However, the absence of real-time radiographic feedback limits the effectiveness of these virtual learning environments [40].

Real-time radiography simulations such as ProjectionVR™ allow for safe, immediate feedback during training, matching the performance improvements seen in traditional settings [7]. Advanced systems address more complex diagnostic scenarios, incorporating animation frameworks and technologies like gVirtualXR (gVXR) for enhanced simulation fidelity [31, 30]. These tools not only provide educational efficiency, but also minimize unnecessary radiation exposure in orthopedic practices by simulating the radiographic process and allowing more precise positioning of equipment using MR technologies [15]. In orthopedic practice, novice radiation technologists often need multiple scout X-ray images to correctly position equipment, exposing both patients and the team to unnecessary radiation. Researchers developed an artificial X-ray imaging system using an optical tracking system and a Digitally Reconstructed Radiograph algorithm for real-time simulated fluoroscopic images in a simulated operating room [33]. Similarly, another team employed MR technology to visualize these simulations which significantly reduced the number of X-ray images required, providing real-time feedback and practical training experience [13]. However, none of these real-time simulations have been adapted to a collaborative MR environment for training purposes.

2.2 Tangible Hybrid Environment

Building on the identified research gap from previous research [40], which highlighted the ethical concerns surrounding repeated sedation of animals and the lack of suitable practice targets in veterinary radiology training, this study advances the field by addressing these

challenges. It suggested incorporating real objects, such as X-ray machines and detectors, to enhance realism and minimize the gap in skill transfer to real-world scenarios [40]. Leveraging this insight, the current study integrates physical equipment with virtual animal models, bridging the gap between simulation and practice while maintaining ethical and practical standards. Tangible interactions in virtual environments enhance realism by allowing users to manipulate physical objects, providing authentic sensory feedback. Using real objects that resemble their virtual counterparts in VR training, as facilitated by the Tangible User Interface (TUI), significantly improves user performance compared to real-world simulations, traditional controllers, and gesture-based interfaces [23]. For instance, a tracked passive haptic ball enhances both performance and presence in comparison to interactions with a purely virtual ball [5]. However, controllers appear more effective due to their superior haptic realism, concentration facilitation, perceived workload, and subjective training success in immersive VR training [23].

In MR, interactions using a tangible cube have been found to enhance user experiences over those involving gestures or mid-air interactions [3, 36], and achieve a comparable level of user experience to controllers (also with haptic feedback). Notably, in MR environments, users must manage tasks in both virtual and real worlds, making constant attention switching mentally challenging, as this dual focus increases cognitive load and impacts user performance [3]. These studies underscore the importance of seamless integration and real-time synchronization between physical and virtual interactions through tangible interfaces, which significantly strengthen the sense of presence and engagement in virtual settings, especially for tasks that mimic real-world activities [4, 35]. Additionally, MR allows learners to practice medical skills effectively and immersively. For instance, researchers implemented a system designed to aid ultrasound practitioners in learning and refining their skills in performing Doppler ultrasounds using a physical 3D abdomen model with an MR HMD [19]. Another study combines the physicality of realistic manikins with the immersive experience of VR to create a training environment [34]. This approach allows trainees to use real tools and engage both physically and verbally with the scenario, enhancing the realism of the training experience. Such setup has huge potential to support various types of Objective Structured Clinical Examinations (OSCE) ¹ with wider possibilities. No OSCE is available for the equine DI procedure, which is one of the problems this research aims to address.

However, some researchers argue that tangible setups do not offer significant advantages over virtual representations in terms of educational effectiveness as the physical setup often requires more time to prepare, posing a potential drawback in educational contexts [16]. It suggests that the effectiveness of tangible interfaces in teaching may depend on the specific task at hand, which is also recommended by other researchers [3]. Moreover, the tangible interface increased participants' awareness of imperfections in projection alignment, which could also heighten the mental load associated with context switching [3]. Based on these observations, one of the objectives of this study is to evaluate their impact on context switching and explore whether tangible setups can be beneficial for veterinary DI procedure training.

3 COLLABORATIVE HYBRID LEARNING ENVIRONMENT

3.1 System Design

An MR system was developed that utilizes spatial sensors to track actual equipment while using virtual horses as a reference for practice [39]. As mentioned in Sec. 1, this system was referred to as PEMR (Fig. 2-A). To evaluate the necessity of using physical

¹The OSCE is a standardized assessment method in medical education that evaluates clinical skills in a controlled environment with multiple stations, each testing different competencies through structured scenarios.

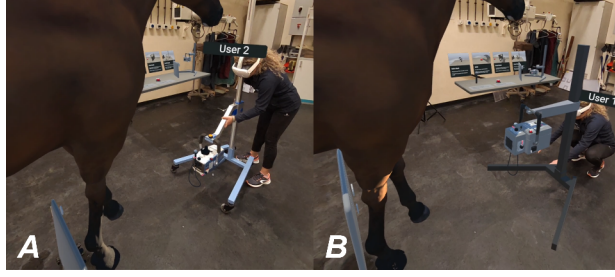


Figure 2: User operating actual X-ray equipment, tracked by VIVE trackers. View from Meta Quest 3 (**Fig. 2-A**). User operating virtual X-ray equipment. View from Meta Quest 3 (**Fig. 2-B**).

equipment for simulating the DI procedure, a system was implemented utilizing only virtual equipment, termed VEMR (**Fig. 2-B**), for comparison with PEMR. Considering radiation safety regulations, the exposure process was digitally simulated, ensuring students' safety when practicing repeatedly. The simulated X-ray images were generated based on real-time location data from the X-ray machine, the horse, and the image detector, allowing students to directly observe the results of their positioning during the training sessions. The system was divided into host and user components: the host component was responsible for tracking the real equipment, sharing this data across the network to all users, and rendering real-time X-ray simulations using a dedicated GPU. The user component involved manipulating actual X-ray equipment and adjusting annotated virtual interfaces in PEMR, or manipulating virtual X-ray equipment in VEMR. Network synchronization ensured that all users were coordinated, receiving task assignments as a group that mirrors real-life scenarios, including the X-ray machine's operation and positioning, and the image detector's positioning with the markers set.

3.2 Devices and SDKs

The user-side equipment within this system was the Meta Quest 3 VR HMDs. The software was developed using Unity Engine (2022.3.14f1) with the [Meta XR Interaction SDK v62](#), which contained Depth API and Spatial Anchor functionalities. On the host side, the system utilized the Legion 5 Pro Laptop, equipped with an NVIDIA GeForce RTX 3060 Laptop GPU, an AMD Ryzen 7 5800H processor with Radeon Graphics, and 32.0 GB of RAM. Additionally, the HTC VIVE SteamVR base station 1.0 and VIVE tracker 3.0 were employed to track X-ray equipment. The host system had two components: a tracking system and a real-time X-ray simulation system. The tracking system was developed with Unity Engine using [Unity OpenVR Tracking \(OVRT\)](#) and the [OpenXR plugin v1.8.2](#). Networking across the system was managed by [Normcore 2](#). The real-time X-ray simulation was executed using Python 3.11.5, and the X-Ray image synthesis was produced using [gVXR](#) open-source library.

3.3 User Interface with High Presence

The proposed system was specifically designed to achieve a high level of local presence within a hybrid reality environment. Rather than limiting interaction to individual users, it supported shared experiences through a network, ensuring the persistence of virtual content. It facilitated interactions not only between a single user and the content (both real and virtual) but also among users, thus mimicking real-world interactions more closely [21]. Moreover, the Depth API allowed virtual content to be occluded by real-world objects and surfaces, making it appear embedded within the physical environment and further enhancing the sense of local presence (**Fig. 3**). Compared to the model-based positioning simulation [17],

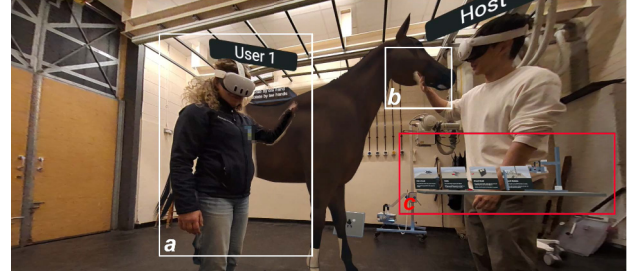


Figure 3: (a, b) Virtual horse with physical occlusion. (c) A table for tutorials on hand-tracking interactions without physical occlusion.

our system implementation retained all essential features, including collimation beam adjustment, visualization, ruler, image detector, marks, and exposure buttons. In PEMR, users were required to manipulate physical equipment in a manner akin to a real clinical setting. The annotations reacted to movements within the physical world, thereby rendering the virtual content world stable and persistent [21]. The hand-tracking feature facilitated a more natural interaction, simplified the system setup, and enhanced the transition between virtual and real contexts, thereby increasing interactivity and potentially enhancing perceived local presence. A tutorial session on hand-tracking was implemented to acquaint users with the virtual content before progressing to practical sessions (**Fig. 3-c**). This session focused solely on basic interactions with the virtual environment, such as pressing buttons and grabbing objects. No instructions or demonstrations related to performing a DI procedure were introduced at this stage.

3.4 Tracking System and Synchronization

As mentioned in [Sec. 3.2](#), the tracking system was maintained on the host system. It monitored all three HTC VIVE trackers using two HTC VIVE SteamVR base stations and shared the transform data with all three Quest 3 devices over the network. Trackers were assigned as follows: one for tracking the X-ray machine, another for tracking the image detector, and a third used as a world anchor for calibration purposes. OVRT was utilized to enable SteamVR tracking for Unity without requiring an actual HMD.

In the host setup, the center of the virtual world was not synchronized with the VIVE tracker system. Typically, the center of the VIVE tracker system corresponded to one of the tracker base stations positioned high up in the room to enhance tracking quality. Therefore, a tracker placed on the ground was utilized to calculate the height and rotation of the base station, aligning the VIVE tracker system with the virtual world.

The synchronization of the user side was achieved by positioning the controller relative to the VIVE tracker physically. Before setting up the virtual environment for the first time, the user placed the controller on the anchor VIVE tracker used for the host side world center and created a spatial anchor that matched the shape and transforms of the anchor VIVE tracker. The spatial anchor could be saved to the Quest 3, storing spatial information related to the current physical environment. Once saved, there was no need to repeat the controller-based calibration during subsequent uses, even if the user accidentally exited the software midway. The spatial anchor could be retrieved and used for future calibration processes.

3.5 Real-time X-ray simulation

gVXR is an open-source framework designed to simulate X-ray images in real time. It leverages the Beer-Lambert law and utilizes GPU to render these images using 3D models (triangular meshes) as foundational elements. In this context, a detailed model could significantly enhance the fidelity of simulated images. For this

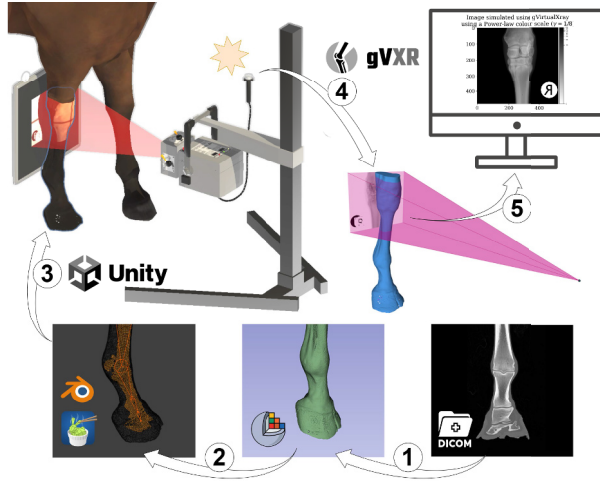


Figure 4: Workflow for creating real-time X-ray simulations: (1) Importing DICOM data into 3D Slicer. (2) Segmenting and generating 3D meshes, then exporting to Blender. (3) Refining and rigging meshes, then importing to Unity and gVXR. (4) Transferring data to gVXR. (5) Synthesizing and displaying virtual X-ray images.

study, Digital Imaging and Communications in Medicine (DICOM) data was used from a horse's right front limb obtained via a Computed Tomography (CT) scan (FRONT LIMB 0.75 I80s 3; Size: 512x512; Count: 1203; SIEMENS SOMATOM scope), sourced from the School of Veterinary Medicine. Using *Slicer 3D* (4.11.0), the skin and bone structures were segmented and exported as 3D models. Notably, the bone was segmented into two distinct parts using different selection thresholds to yield more accurate outputs (Fig. 4-1). Subsequently, the skin model underwent retopology using the open-source tool *Instant Meshes* [12] to further refine the mesh topology, as the details of the flesh were not clearly visible in X-ray imaging. Additionally, the model was simplified and fully rigged using *Blender* (v4.0.1) (Fig. 4-2), because the limb's CT scan was not positioned in a standing pose, which could give inaccurate results in the simulation for this study if placed directly on top of the low-poly horse model. Rigging the model ensured that it was displayed in a standing position, aligning more effectively with the low-poly horse model in Unity (Fig. 4-3).

When the exposure button was triggered, the data required to generate the X-ray simulation including the X-ray source, target object, X-ray detector, collimation size, and markers would be transferred to gVXR (Fig. 4-4). The data was transferred from the host Unity end to the Python end using a low-level networking interface socket through 127.0.0.1 localhost. The transform data required to generate the X-ray simulation included the X-ray source, target object, X-ray detector, and markers. Since the gVXR did not offer functions to rotate or set the rotation of the X-ray source, all other context transform data was calculated relative to the X-ray machine's transform in the Unity end before being sent to the Python end. Furthermore, there was a difference in coordinates between Unity and the gVXR OpenGL environment, where Unity used XYZ (with the y-axis pointing up), while gVXR used XZY (with the z-axis pointing up). Therefore, coordination transformation must be taken into account when inputting parameters. Additionally, the number of pixels and size of the detector in gVXR were necessary to determine the size of the final output. In the Unity end, the size was calculated using raycasting from the X-ray source to the X-ray detector.

gVXR imported all necessary models while configuring the model's elements, compounds, densities, and mixtures. Upon re-

ceiving data from the user, it started the X-ray simulation: initializing the X-ray source, configuring the image detector, adjusting the model transformations, and creating the simulated X-ray image (Fig. 4-4). The Python end first created the OpenGL environment and loaded all the required models into the environment. Simultaneously, the elements, compounds, densities, and mixtures of the model were set up accordingly. Subsequently, it consistently listened to the localhost. Once the exposure button was triggered from the user side, the host Unity end received the command and sent all the required transform data to the Python code. Then, Python begins calling the gVXR functions to generate the X-ray simulation. The procedure was as follows: (i) It set the source position to idle, the source type to a point source, and a monochromatic beam (in the current system, 1000 photons of 80 keV were set). (ii) It configured the image detector's position, up vector, and right vector, and determined the number of pixels and pixel size (in the current system, the pixel size was set to 0.5 mm). (iii) The models and markers were translated to their relative position and rotated to their relative rotation to the X-ray source. (iv) The X-ray image was simulated, visualized, and displayed. After generating the radiograph, users could review the output on the host display, which replicated the actual procedure used in a DI room.

4 EXPERIMENT METHOD

4.1 Study Setting and Participant

This user-centered comparative study employed a within-group design, with the independent variable consisting of two primary conditions. In the first condition (physical equipment), participants interact with PEMR, while the second condition (virtual equipment) involves VEMR. The dependent variables are the students' performance, measured by their scores and the time taken to complete a DI procedure, and their behavior measured by their movement data. Participants were randomly assigned to one of two groups: one group started with PEMR and then switched to VEMR, while the other group did the reverse. This design allowed participants to experience both systems while mitigating knowledge carryover and reducing fatigue accumulation during the sessions. This study was reviewed and approved by a recognized ethics review board. The Office of Research Ethics at University College Dublin granted an exemption from a full ethical review under protocol number LS-C-22-183-Kilroy (originally approved on 14 September 2022, with an extension approved on 18 April 2024).

Twenty-one veterinary students enrolled in a weekly clinical rotation consented to participate in the study. They were recruited

Table 1: Demographics of participants in this study (N = 24)

Characteristic	PEMR first	VEMR first
Role in Veterinary Medicine field		
Final year Veterinary student	10	10
Diagnostic Imaging expert	2	2
Gender		
Male : Female : Prefer-not-to-say	2:8:2	2:10:0
Dominant hand		
Right : Left : Both	9:3:0	12:0:0
Age range		
21 - 25 : 26 - 30 : 31 and above	7:2:3	7:3:2
Experience with VR for gaming		
Never : Once : Occasionally	5:5:2	9:3:0
Median hours of video game experience each week (range, IQR width)	0 (0-5, 0)	0 (0-1, 0)

* IQR = interquartile range.

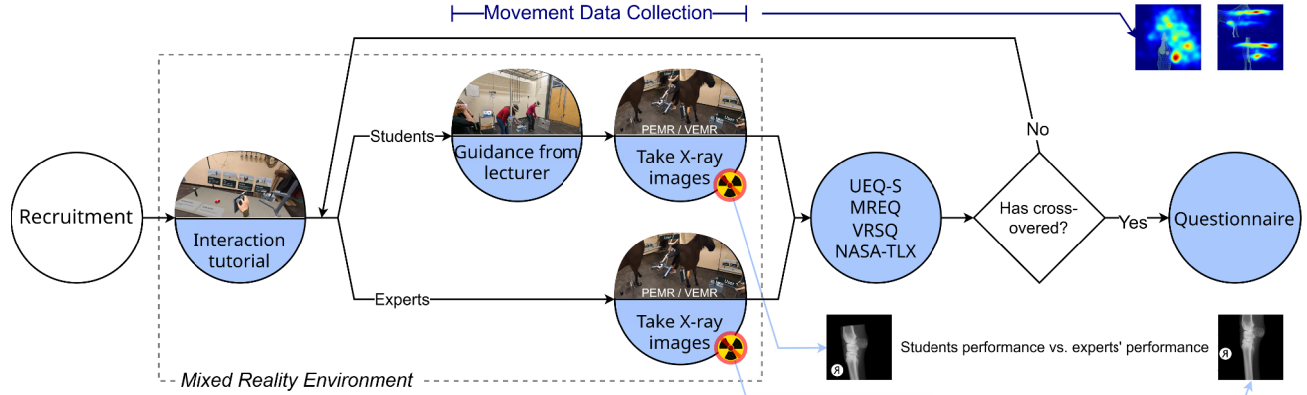


Figure 5: Study schema illustrating the different procedures taken by students and experts, and the times at which movement data and X-ray image data were collected.

weekly by the radiologist involved in the study, and all knew each other, as they were part of the same final-year class. However, one student withdrew due to personal circumstances. Additionally, four DI experts were recruited to establish a benchmark: a resident with six years of experience, a radiographer with twenty-one years of experience, a radiologist with seventeen years of experience, and a diagnostic imager with four years of experience. Consequently, the study included 20 veterinary students and four DI experts. Participants provided demographic information, including their experience with VR, and the amount of time they spent gaming each week (Tab. 1). Most participants reported limited VR gaming experience and minimal weekly video game engagement.

4.2 Procedure and Task

The procedure is illustrated in Fig. 5. Before the sessions, participants were required to sign a consent form and complete an anonymous demographic questionnaire, ensuring that responses throughout the study were contextualized without revealing individual identities. Then, they were asked to complete a hand-tracking interaction tutorial. This tutorial included a mini version of the virtual X-ray machine and image detector and demonstrated basic operations such as palm grabbing, pinch grabbing, and poking, ensuring participants knew how to interact with the virtual elements.

Participants started with a session randomly allocated to either the PEMR or VEMR, determined by the experimenter to maintain a balanced group distribution. In the practical phase, participants, working in pairs, used an X-ray machine and an image detector to perform two X-ray procedures on a specific area of a virtual horse's right front limb. For each procedure, one participant held the image detector while the other operated the X-ray machine. After the first X-ray, roles were swapped for the second, and so on. Students received direct instructions and guidance from an on-site lecturer with an HMD. The radiologist demonstrated the same X-ray scan techniques and provided additional information based on students' questions, similar to a typical teaching scenario. The experts completed the process without these aids, and their performance served as a benchmark to evaluate the students' performance.

While the technical experimenter prepared for the second session, participants were asked to put down the HMD and complete a brief questionnaire that included several user experience questionnaires, which will be detailed in Sec. 4.4. The second session mirrored the first but used an alternative system to facilitate comparative analysis. Participants once again used the HMD and repeated the X-ray procedure with their group mates under the guidance from the on-site lecturer. To conclude, they filled out the

same questionnaire as before, along with a questionnaire containing some open-ended questions comparing the two different systems and gathering their feedback and suggestions.

4.3 Material

During the practical phase, movement data from on-site users, X-ray machine, and image detector would be collected every 0.5 seconds (Fig. 5). The data would be stored in JSON format, including frame, timestamp, user name, and position, with the position data calculated relative to the position of the horse's right front limb. Heat maps would be generated to analyze user behavior, providing insights into areas of interest and user engagement for both systems, clearly highlighting the similarities and differences between the two conditions. Additionally, the time participants spent in MR would also be recorded. To assess participants' performance in taking X-ray images, data were recorded on two main aspects during each exposure action: radiation safety and image quality. The distances between each user and X-ray machine ($D_{U,X}$) and image detector ($D_{U,ID}$) were recorded as the further the users stayed away from the X-ray beam, the less chance they got radiated. Distance scores ($DS_{U,X}$ and $DS_{U,ID}$) were calculated by assigning values from 0 to 4 (distinct) based on ranges of $D_{U,X}$ and $D_{U,ID}$, which determine the score of radiation safety (S_{RS}).

The image quality score was determined by the following factors: whether there was a beam leakage² (V_B); whether the marker was correctly placed (V_M); whether the target area was centered (V_C); the distances between X-ray machine and target ($D_{XM,H}$) and image detector ($D_{XM,ID}$); the alignment of the image detector to the ground level (A_G) and the X-ray machine (A_{XM}). Again, distance score and angle score, $DS_{XM,H}$, $DS_{XM,ID}$, $AS_{IM,G}$, and $AS_{IM,XM}$, were evaluated with scores ranging from 0 to 4, which were then integrated to calculate the score of image quality (S_{IQ}).

The overall score combines S_{RS} and S_{IQ} using a weighted average based on the number of components, where S_{RS} contributes 30% and S_{IQ} contributes 70% to the assessment. The time spent on each X-ray was also recorded, starting from when the lecturer assigned the task until the X-ray image simulation was generated.

4.4 Questionnaire

The questionnaire was designed based on the short version of the User Experience Questionnaire (UEQ-S) [27], the NASA Task

²Beam leakage occurs when the primary beam extends beyond the collimation boundaries, reaching areas outside the region of interest or the image detector. This indicates poor practice, as it can expose nearby personnel to primary radiation, raising radiation protection concerns.

Load Index (NASA-TLX) [11], the Mixed Reality Experience Questionnaire (MREQ) [22], and the Virtual Reality Sickness Questionnaire (VRSQ) [14]. The MREQ, adapted to fit the system setup, was used to assess the participants' sense of presence in the MR environment. For each system, 8 out of 33 items were applied to measure the sense of presence. The full versions of UEQ-S, NASA-TLX, and VRSQ were employed to evaluate perceived user experience, task load, and potential discomfort, respectively. In addition to the Likert-scale questionnaires, several open-ended questions were included to gather their feedback and suggestions.

4.5 Statistical Analysis

The main hypotheses regarding the study were: (H1a) Students' X-ray image quality will improve after the practical session. (H1b) Students' X-ray image acquisition time will improve after the practical session. (H2a) Experts will outperform students' image quality. (H2b) Experts will outperform students' image acquisition time. (H3a) Participants will prefer PEMR to VEMR for user experience and classroom integration. (H3b) Participants will feel more present in PEMR than in VEMR.

Simple descriptive statistics from the survey were presented. The UEQ data were analyzed using the Excel data analyzing tool, provided by the UEQ team. The VRSQ score was calculated for the oculomotor and disorientation components. Regarding the score and time for taking the first and last radiography in PEMR and VEMR systems by students, the Shapiro test assessed the normality. Outliers were identified using box-and-whisker plots. For normally distributed data, a one-tailed paired t-test was conducted to determine whether there were significant differences. For non-normally distributed data, a Wilcoxon signed-rank test was conducted. These data were also tested against the mean scores achieved by experts and the mean time they took to perform one DI procedure as benchmarks in both systems using a one-sample t-test. A p-value of < 0.05 was considered statistically significant, and exact p-values were reported except for those that were < 0.001 . Significance values were reported in brackets for $p < 0.05$ (*), $p < 0.01$ (**), $p < 0.001$ (***), and $p < 0.0001$ (****) respectively. Data analysis and charts were performed using Python, with packages including matplotlib, pandas, seaborn, pingouin, and numpy.

5 RESULT

5.1 X-ray Image Score and Time Spent

The scores and the time spent taking X-rays are shown in Fig. 6. In general, students performed significantly better in VEMR than in PEMR, either when taking the first X-ray ($\bar{x} = 78.14\%$, $\sigma = 8.64\%$ vs. $\bar{x} = 58.36\%$, $\sigma = 11.09\%$; ****, Wilcoxon signed-rank test) or the last one ($\bar{x} = 77.39\%$, $\sigma = 13.02\%$ vs. $\bar{x} = 67.95\%$, $\sigma = 10.23\%$; $p = 0.016$, paired t-test). Moreover, their performance in VEMR showed no significant difference from the experts' performance (vs. $\bar{x} = 77.85\%$; $p = 0.15$ and $p = 0.16$, one-sample t-test). The first image and the last image showed no significant difference using VEMR ($\bar{x} = 78.14\%$, $\sigma = 8.64\%$ vs. $\bar{x} = 77.39\%$, $\sigma = 13.02\%$; $p = 0.40$, paired t-test), while there was a significant improvement after the session in PEMR ($\bar{x} = 58.36\%$, $\sigma = 11.09\%$ vs. $\bar{x} = 67.95\%$, $\sigma = 10.23\%$; ****, paired t-test). The students' first attempt in PEMR scored significantly lower than the experts' ($\bar{x} = 58.36\%$, $\sigma = 11.09\%$ vs. $\bar{x} = 67.10\%$; $p = 0.0011$, one-sample t-test). After the practical session, there was no significant difference in the performance of the two groups ($\bar{x} = 67.95\%$, $\sigma = 10.23\%$ vs. $\bar{x} = 67.10\%$; $p = 0.357$, one-sample t-test).

Students spent significantly less time taking the last X-ray image using PEMR than with VEMR ($\bar{x} = 289.1$ s, $\sigma = 64.56$ s vs. $\bar{x} = 335.7$ s, $\sigma = 129.30$ s; $p = 0.047$, paired t-test). Apart from this, no significant differences were observed, either between the time taken to obtain the first X-ray image in PEMR ($\bar{x} = 305.8$ s, $\sigma = 92.02$ s) and VEMR ($\bar{x} = 317.0$ s, $\sigma = 87.07$ s), or the time taken to acquire

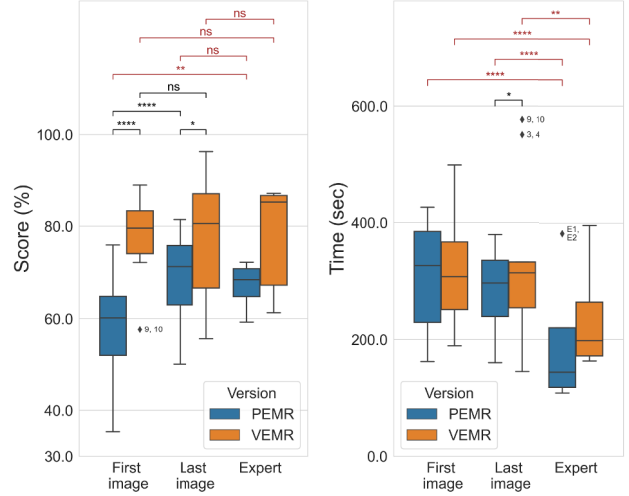


Figure 6: Box plot of the scores and time spent for the first and last X-ray images during the session, and the expert's X-ray image score and time spent.

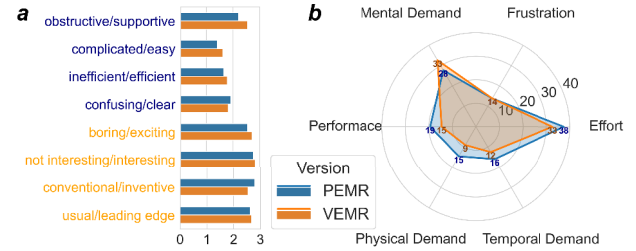


Figure 7: Result of UEQ-S. The range of the scales is between -3 (very bad) and $+3$ (extremely good). Pragmatic Quality scale and Hedonic Quality scale (Fig. 7-a). The radar chart of NASA-TLX scores for each workload sub-scale. The lower an index is, the more demanding (worse performing) the user finds the task (Fig. 7-b).

the first and the last image in both systems. In PEMR, the experts took an average of 194.5 seconds, while in VEMR, the average was 238.8 seconds. Throughout the practical sessions in both systems, the students consistently spent significantly more time than the experts on both the first (PEMR: ****; VEMR: ****) and the last images taken (PEMR: ****; VEMR: $p = 0.0017$).

5.2 User Experience

The results of UEQ-S for each sub-scale are shown in Fig. 7-a. The pragmatic quality of PEMR is 1.802 ($\sigma = 0.906$) and the hedonic quality is 2.677 ($\sigma = 0.503$). The overall score for PEMR is 2.240 ($\sigma = 0.663$). The pragmatic quality of VEMR is 1.948 ($\sigma = 0.921$) and the hedonic quality is 2.688 ($\sigma = 0.523$). The overall score for VEMR is 2.318 ($\sigma = 0.662$). The UEQ offers a benchmark that contains the data of 468 product evaluations (21,175 participants), with the UEQ. Compared to the benchmark, both systems reached the range of the top 10% of results. No significant differences were found between the two systems across all sub-scales in UEQ-S. The NASA-TLX questionnaire measures perceived workload. As shown in the radar chart (Fig. 7-b), VEMR was perceived as more mentally challenging, whereas PEMR required more physical effort. Participants believed they achieved better outcomes more effortlessly and with less time pressure using VEMR compared to PEMR. However, their level of frustration remained con-

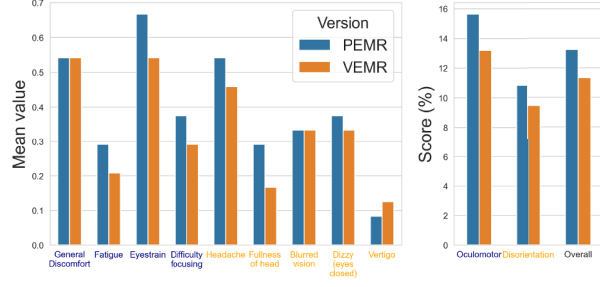


Figure 8: Result of VRSQ. Oculomotor scale and Disorientation scale. Mean scale: 0 = None, 3 = Severe.

sistent across both systems. No significant differences were found between the two systems across all sub-scales.

The MREQ is designed to measure users' sense of presence and their general experienced perception of an MR environment [22]. An eight-item subset was applied from the full MREQ (PEMR: $\bar{x} = 6.672$, $\sigma = 0.695$; VEMR: $\bar{x} = 6.792$, $\sigma = 0.467$) as shown in Tab. 2. Each item is rated from one to seven. In this hybrid reality environment, [agents] referred to the lecturer and other users who were present in the real environment; [real environment] referred to the actual Equine DI room where the radiographer takes radiography from big live animals like horses; [real objects] referred to the real X-ray machine, and the real image detector in PEMR; [virtual objects] referred to the virtual horse, annotations on virtual X-ray machine/X-ray machine, and virtual image detector. The means are significantly above the high point (6.0) picked as the benchmark (PEMR: ***, one-sample t-test; VEMR: ***, one-sample t-test). No significant differences were found between the two systems across all sub-scales in MREQ.

The average HMD wearing time is 11.41 minutes per session. The results of the VRSQ are shown in Fig. 8, with PEMR scoring

Table 2: Result of MREQ

Relation	Item(s)	PEMR ^a	VEMR ^a
Usr-RE	I felt I was part of the room.	6.625	6.667
Usr-RO/ Usr-VO	The horse, the X-ray machine, and I were in the same environment.	6.417	6.750
Usr-RA	The lecturer, other students, and I were in the same room.	6.792	6.875
RO-RE/ VO-RE	The horse and X-ray machine belonged to the room.	6.542	6.708
RO-VO/ VO-VO	The horse and X-ray machine were part of the same space.	6.625	6.833
RO-RA/ VO-RA	The lecturer and other students in the environment were in the same space as the horse and X-ray machine.	6.750	6.792
RE-RA	The lecturer and other students were in the room.	6.792	6.792
RA-RA	The lecturer and other students could have communicated with each other.	6.833	6.917

*Usr = User, RE = real environment, RO = real object, VO = virtual object, RA = real agents

^a Mean value: 1 = Strongly disagree, 7 = Strongly agree.

13.23 and VEMR scoring 11.32 (ranging from 0 to 100), based on Kim et al. [14]. Generally, users experienced more symptoms in the oculomotor and disorientation scales after using PEMR. However, no significant differences were found between the two systems.

5.3 Qualitative Observation

5.3.1 Preference and Rationale

Participants answered open-ended questions on differences between the PEMR and VEMR systems, overall feedback, and suggestions. Fourteen of the 24 participants, three experts among them, preferred VEMR. They thought "it was easier to use and understand," (P2³) and "flowed more seamlessly and felt more natural" (P6). Compared to PEMR, it was less "trippy between what was real or not" (P10). They also preferred the fact that they did not need to lift the real equipment (P12). Meanwhile, 9 out of 24 participants preferred PEMR, while one expert liked them equally. They stated that PEMR felt more realistic (P4) and "it was nice to learn while feeling like we were using the actual equipment" (P20). Some of the participants who preferred PEMR thought it was "easier to control" (P7) when compared with VEMR.

5.3.2 Realism and Adoption in Training

Twenty of the 24 participants, three experts among them, agreed that PEMR better simulated a real-world scenario, as "there was the actual plate" (P2) and they controlled "the actual movement of the machine and plate" (E3). Meanwhile, the remaining four participants thought VEMR was closer to a real-world scenario. When asked about which version they would accept as a regular training tool in their teaching routine, 7 out of 24 participants accepted VEMR, while another 7 accepted PEMR. Ten out of 24 participants accepted both systems as regular teaching tools. One of these participants preferred VEMR, noting that it "would be simpler and more efficient to carry out" (P17).

5.3.3 Necessity of Real Equipment

Sixteen of the 24 participants, two experts among them, agreed on the necessity of using physical equipment. They argued that real equipment enhanced realism and helped students learn to maneuver it, as one stated, "a large part of the learning process is getting comfortable with and handling the equipment, adjusting height, etc., even getting used to the weight of holding the plate" (P18). They especially noted that the weight of the equipment enhanced realism, as "you have to center [the radiation beam] correctly whereas [in VEMR the image detector] can be put anywhere" (P16). Meanwhile, one participant provided feedback suggesting that "maybe not as a VR activity - could do that after with just a horse model to practice movement of the equipment" (P15). Lastly, 7 out of 24 participants did not think physical equipment was necessary, as the virtual equipment "worked just as efficiently" (E2) and was "less strenuous" (P12). Some found PEMR difficult to navigate (P6) and more confusing (P9).

5.3.4 Portability vs. Physical Interaction

Eleven of the 24 participants, one expert among them, believed VEMR's portability is more beneficial than physical interaction in the PEMR, as stated "[VEMR] is good if it can be brought around" (P11). Meanwhile, 8 out of 24 participants, one expert among them, felt that physical interaction is more important as it simulates a real-life scenario (P16). Lastly, 5 out of 24 participants, two experts among them, thought that both systems were good learning options. They suggested that the corresponding system could be used based on the available teaching resources, as stated "both [systems] have their advantages and both will be a good way to learn - if a portable

³P refers to participant; E refers to expert.

one is needed to teach students, it will be better, but if there is equipment available, that may be better for teaching as you are able to move around the actual equipment” (P17).

6 DISCUSSION

6.1 DI Training with Real-time Feedback

The system enabled multiple students to collaborate in taking X-rays of a virtual horse. Compared to a real equine patient, which is large and has unpredictable temperaments, the virtual patient is easier to maneuver and safer for on-site students. Additionally, the X-ray image, based on the actual equipment, is generated in real-time, allowing learners to immediately confirm their learning outcomes.

Under real-time guidance from an on-site lecturer who also wore an HMD, the students’ performance (image quality and safety rule) improved significantly after the practical session in PEMR. This result indicates that a system utilizing an actual X-ray machine and image detector can be a supportive teaching tool for equine diagnostic imaging technique training. In this case, H1a was supported in the PEMR setup. However, students’ performance remained consistent in VEMR throughout the practical session and was significantly better than their performance in PEMR, leading to H1a (Students’ X-ray image quality will improve) being rejected in VEMR. Handling physical equipment is more complex than it seems, as one radiologist holds the heavy image detector while the other adjusts the X-ray machine. Multiple adjustments can make holding an image detector physically demanding and strength-intensive. Compared to using physical equipment, manipulating virtual equipment is relatively easier since it can float mid-air without consistent handling. With the lecturer’s guidance, students can slowly adjust equipment positioning from any angle, unrestricted by real objects, resulting in better performance. This outcome aligns with the participants’ feedback as most of them stated the virtual equipment is easier to handle, while some of them held opposite opinions.

There was no significant improvement in the time students spent taking X-rays after the practical session in either system, leading to the rejection of H1b (Students’ X-ray image acquisition time will improve) for both. Additionally, there was no significant difference in the time required to take X-ray images between the PEMR and VEMR systems for the first image taken. While a significant difference was observed in the last image taken, the difference (an average of 46 seconds) was not substantial. Based on the data, students appeared to gain experience in positioning more quickly in PEMR than in VEMR.

Both systems received high praise from participants, as noted: *“Amazing experience! I believe this will be an invaluable tool for students to gain a better understanding of DI before working with a real animal, which greatly enhances their knowledge and safety.”* (P4) The systems offered a high user experience, with very high hedonic quality and mid-to-high pragmatic quality, as well as a low task load index. Although VEMR received a relatively high average value, no significant difference was observed, indicating that replacing physical equipment with virtual equipment does not significantly impact user experience.

6.2 Using MR Techniques for Assessment

Four DI experts were invited to test both systems and evaluate their performance, using their scores as benchmarks for comparison with the student performances. This assessment was intended to determine whether the systems could serve as potential tools for DI training. Since the current DI institution lacks a standardized assessment method, an initial assessment rule (described in Sec. 4.3) was created by calculating scores based on the user’s standing position, the X-ray equipment’s positioning, and the quality of the X-ray images, following recommended principles and lecturer suggestions. In PEMR, students initially performed significantly worse than experts on their first image, even under the supervision of the on-site

lecturer. However, their final images showed no significant difference from those of the experts, indicating that with on-site guidance and practice using PEMR, students could achieve an expert level of performance. Thus, hypothesis H2a (Experts will outperform students’ image quality) was rejected for PEMR. In VEMR, students’ performance showed no significant difference from that of the experts throughout the practical session, so H2a was also rejected for VEMR. Regarding the time taken to capture each X-ray image, the experts consistently spent significantly less time than the students across both systems, confirming hypothesis H2b (Experts will outperform students’ image acquisition time) in the PEMR and VEMR systems. Both experts and the students who participated in the study had little experience with MR technologies. Experts spent 102.95 seconds less than students using real equipment and 87.6 seconds less with virtual equipment, indicating a significant performance gap between experts and students, even with guidance from an on-site lecturer. This demonstrates the potential of using the system to evaluate students’ learning outcomes.

As mentioned in Sec. 1, the DI institution does not have a standardized method for assessment (or even a consistent practical teaching approach). Consequently, students only acquire theoretical knowledge without sufficient opportunities to practice their skills. The performance difference between using real equipment and virtual simulations underscores the crucial role of real equipment in facilitating motor learning and achieving more realistic learning outcomes. Conversely, if real equipment is unavailable for practice, the use of virtual equipment could be considered for an interactive multiple-choice question to test each student’s knowledge.

6.3 The Necessity of Physical Equipment in MR

As mentioned in Sec. 2.2, while tangible setups are innovative educational approaches, their effectiveness compared to virtual representations remains a topic of debate [16]. Research indicates that the suitability of tangible interfaces for teaching largely depends on the specific task at hand. This study serves as an introduction to evaluating the necessity of physical equipment by comparing it with virtual equipment in equine radiography technique training, potentially representing a general training scenario involving collaborative equipment positioning and manipulation. Both systems have unique advantages that the other cannot offer. VEMR is more portable, as it does not require real equipment, while PEMR allows for tangible interaction with physical equipment. In this section, the discussion will be carried out by comparing the differences between the two systems.

6.3.1 Realism Improvement and Equipment Manipulation

The necessity of physical equipment is emphasized for providing realism and familiarizing users with equipment handling, weight, and positioning. Participants considered PEMR to more accurately simulate real-world scenarios due to the use of actual equipment and the ability to control the machine and plate directly. This was reflected in their performance (Fig. 6), which was significantly worse when using real equipment compared to virtual equipment. When it comes to training, the challenges of working with real objects must be considered. For instance, when positioning the real image detector, the user has to hold the handle and stand away from the radiation area. In this stance, it is challenging to verify whether the image detector is correctly positioned and to move the marker without putting the detector down. Therefore, the image detector holder relies on guidance and support from teammates. Using virtual equipment, the situation can be easily managed by one person. Thus, physical equipment is crucial for realistic maneuvering.

Virtual equipment is simpler and more efficient for regular training. Despite this, some believe virtual equipment is equally effective while being less physically demanding.

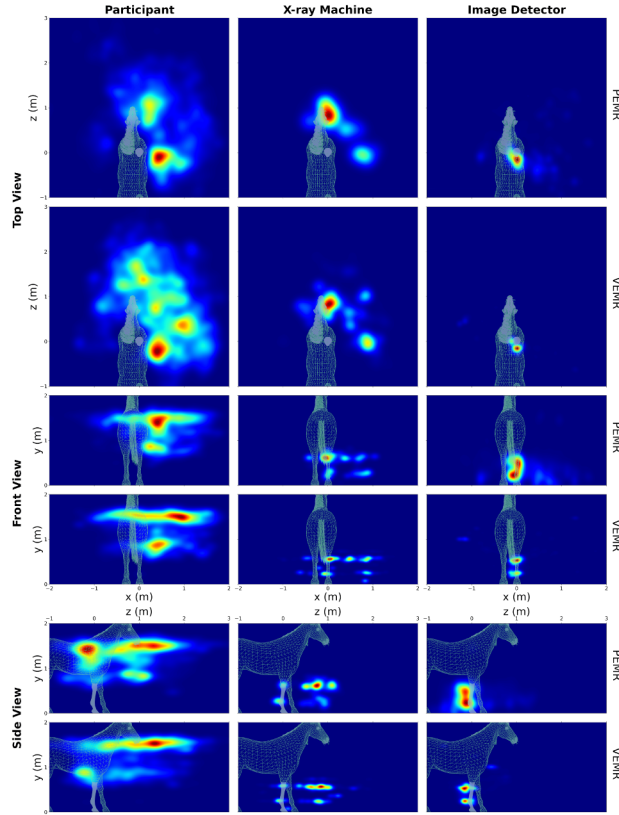


Figure 9: Heat maps of participants', X-ray machine's, and image detector's movements around the virtual horse.

Preferences for these systems and adopting them as regular teaching tools are evenly divided, hence, H3a (Participants will prefer PEMR to VEMR) is rejected. There is consensus that incorporating both approaches could provide comprehensive training. The virtual equipment can be used to enhance the theory learning, while the real equipment is essential for a more close-to-reality learning. However, it is also important to mention that physical devices require considerable preparation time and equipment availability, which can limit practicality in some educational settings. Therefore, implementing tangible setups should be done thoughtfully, ensuring they are tailored to meet the specific requirements of each educational scenario [3].

6.3.2 Behavior in Hybrid Virtual Environment

As shown in Fig. 9, top-view data indicates that users spent most of their time around the X-ray machine and image detector, generally moving between tasks assigned to them during the practical session. Moreover, they tended to move more when operating the virtual equipment than the real equipment. This finding is interesting because, when real obstacles were involved, users tended to maintain their "territory" without moving much. Furthermore, users would not intentionally collide with the virtual horse, which demonstrated its strong presence. This result is supported by the MREQ questionnaire, which aligns with previous research [40].

Although the X-ray machine and the detector showed similar movement patterns in both systems, the real equipment had a wider movement area around the movement hotspots, indicating that more small adjustments were made when positioning. This phenomenon is even more apparent when observing from the side and front views of the heat maps: with the virtual equipment's precise positioning,

the movement with the real equipment was more consistent and broader. Given the real detector's weight, holding it in the air requires some effort. Overall, the virtual equipment movement followed the same pattern as the real equipment, demonstrating that the virtual simulation can replicate the real equipment to a fine degree, but cannot be a replacement for the real equipment. Hence, to bridge the gap between virtual training and real-world scenarios, real equipment is essential.

6.3.3 Potential Risk of Contextual Switch

Although handling real objects seems more instinctive, some participants still found navigation challenging and confusing. Additionally, tangible interfaces can increase users' awareness of projection alignment issues, potentially raising cognitive load due to context switching [3]. Despite the low tracking quality, the combination of real and virtual elements at close viewing distances might cause confusion, as at least one participant reported feeling disoriented between what was real or virtual in PEMR.

6.4 Limitation and Future Work

In a previous pilot test, the students were allowed to take radiographs independently. The results showed that students were unable to successfully perform even a single X-ray without on-site guidance. Due to time constraints for using the radiation room, this process was shortened by only allowing the students to perform the DI technique under the supervision of an on-site lecturer. As a result, data assessing students' previous knowledge levels was not collected, limiting the evidence for using the system for assessment. Moreover, network quality is closely tied to signal strength, which could lead to internet interruptions during the session. In that case, both users and hosts must reset and resynchronize the virtual environment, posing a significant drawback for its adaptation to real training procedures. In addition, the tracking quality is also a concern, as participants reported that the virtual annotation was sometimes not aligned with the movement of the equipment. This misalignment might affect participants' preferences for non-tangible setups. The system will be evaluated through further experiments. Veterinary students scheduled for clinical rotations will be recruited for a longitudinal study to assess the system's effectiveness in facilitating their understanding of radiographic techniques and safety protocols. Additionally, the system's potential for practical assessment will be examined by comparing the performance of novice trainees with no guidance with that of radiography specialists, exploring its possible use for OSCEs and similar evaluations.

7 CONCLUSION

Although simulation cannot truly replace real-world training, its inclusion as demonstrated by this research, shows how it could become integral to the future of veterinary DI training programs. MR will allow some components of training to potentially begin virtually and then progress to tangible MR training, minimizing radiation exposure and eliminating the need for a real horse, while still providing hands-on experience. This approach reduces training logistics, allows more students to be taught, cuts costs, and has potential applications in other fields using real-world equipment. This work has illustrated this future vision by comparing physical and virtual equipment for veterinary DI training in a proposed MR system, demonstrating that virtual objects are effective and less demanding for beginners. However, real equipment is essential to bridge the gap to real-world scenarios, and a combination of both real and virtual equipment would ultimately allow more veterinary students to be trained overall.

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